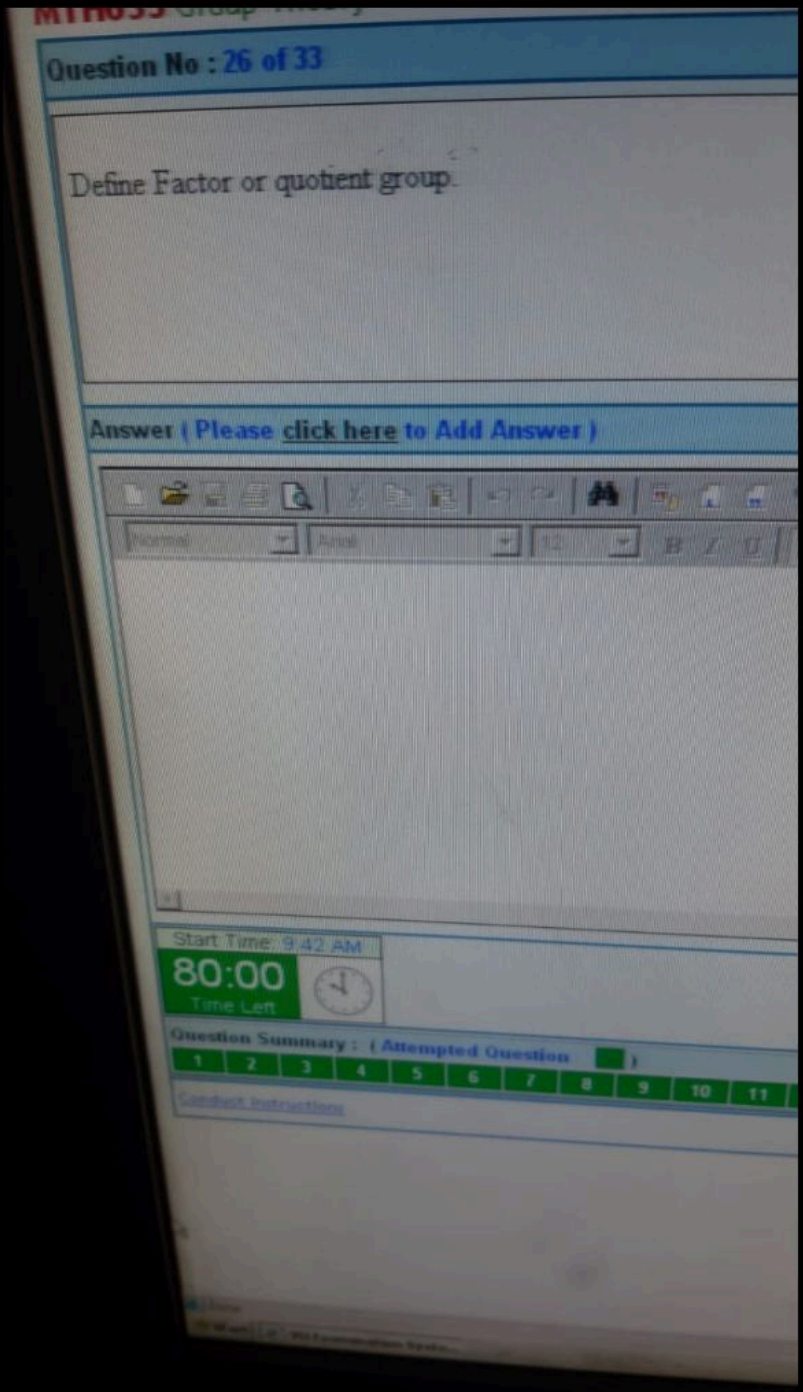
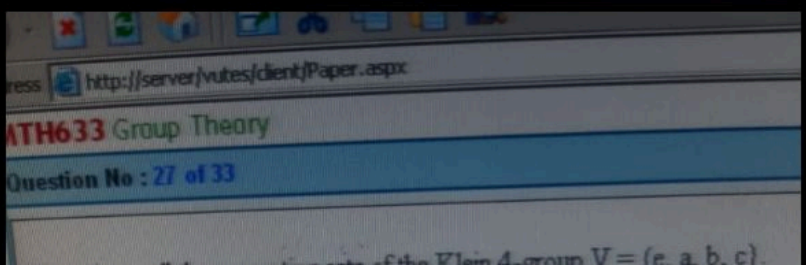

Question No : 26 of 33
Define Factor or quotient group.
Answer (Please click here to Add Answer)
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TH633 Group Theory
Question No : 27 of 33

Factor Group

Definition

Let $(H,*)$ be a normal subgroup of the group $(G,*)$. $(G/H,\otimes)$ is called quotient group, where the operation $\otimes$ is defined on G/H by

$Hg_1 \otimes Hg_2 = H(g_1 * g_2)$.

If G is a finite group, then G/H is also a finite group, and $|G/H| = |G|/|H|$.

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ATH633 Group Theory

Question No : 27 of 33

Write down all the generating sets of the Klein 4-group $V = (e, a, b, c)$.

Answer (Please click here to Add Answer)

Generating Sets

Example

The Klein 4-group V = {e, a, b, c} is generated by {a,b} since ab=c.

It is also generated by {a,c}, {b,c }, and {a,b,c}.

If a group G is generated by a subset S, then every subset of G containing S generates G.

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MTH633 Group Theory

Question No : 32 of 33

Show that the homomorphism $h : G \to G'$ is injective if and only if $Ker\ h=(e)$.

Answer (Please click here to Add Answer)

100%

Injective Homomorphism

Theorem

A homomorphism
h: G→G' is injective
if and only if
Ker h={e}.

Kernel of an Injective Homomorphism

Ans

Proof

Suppose h is injective,
and let x ∈ Ker h.
Then h(x)=e'=h(e).
Hence x=e.

Injective Homomorphism

Conversely, suppose
Ker h={e}.
Then h(x)=h(y)
$\Rightarrow h(xy^{-1})=h(x)h(y^{-1})$
$=h(x)h(y)^{-1}=e'$
$\Rightarrow xy^{-1} \in$ Ker h
$\Rightarrow xy^{-1}=e$
$\Rightarrow x=y.$
Hence, h is injective.

Group Theory

Factor Groups from Normal Subgroups

ATH633 Group Theory

Question No : 30 of 33

Let G be a group. The set of all commutators $aba^{-1}b^{-1}$ for $a, b \in G$ generates a subgroup C of G

Answer (Please click here to Add Answer)

The Commutator Subgroup

Proof

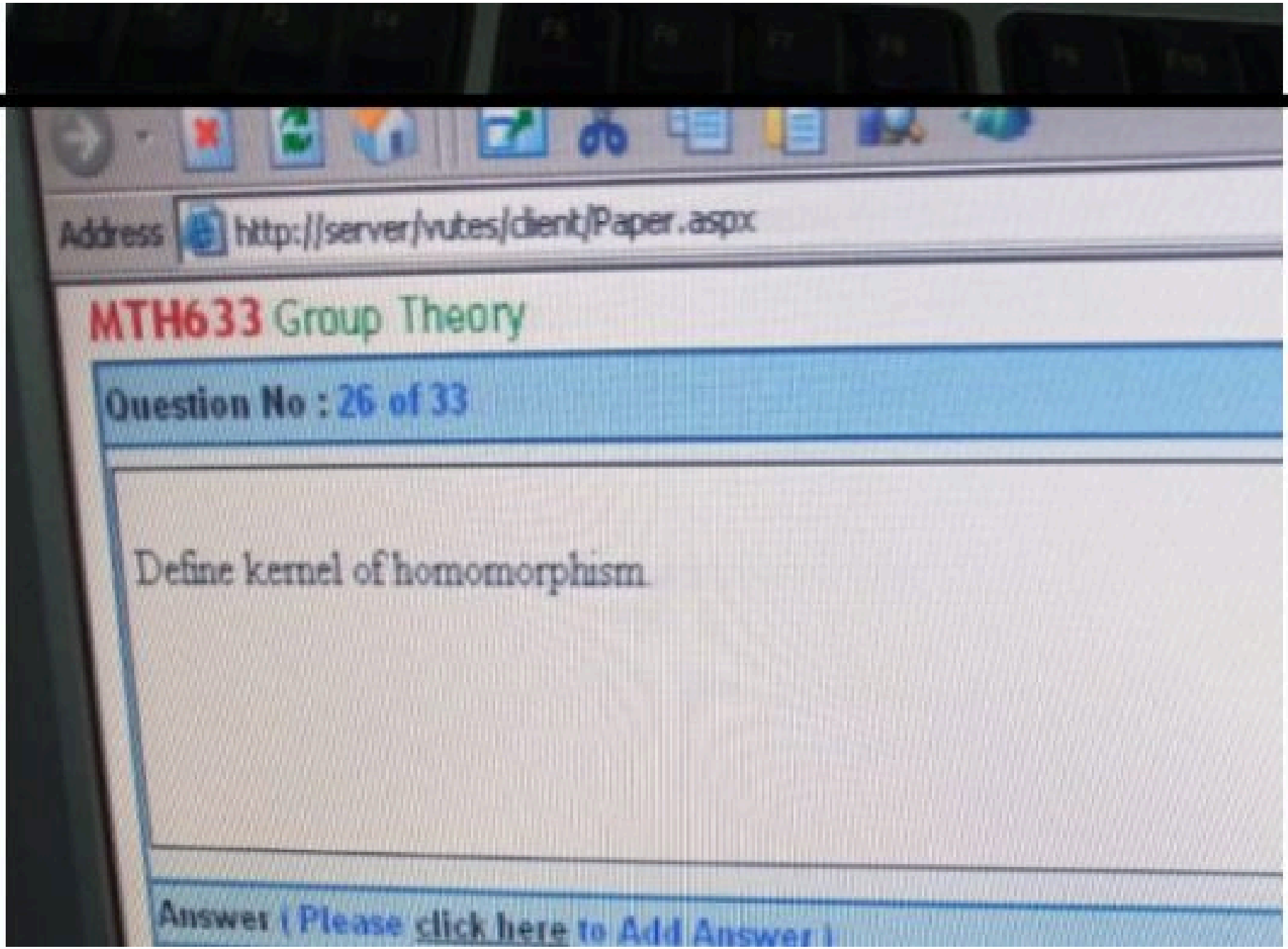
Let $a, b \in G$. Then,

$(aba^{-1}b^{-1})(aba^{-1}b^{-1})^{-1}$

$=aba^{-1}b^{-1}bab^{-1}a^{-1}$

$=e \in C$

since $e = eee^{-1}e^{-1}$ is a commutator.


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MTH633 Group Theory
Question No : 26 of 33
Define kernel of homomorphism
Answer (Please click here to Add Answer)

Examples on Kernel of a Homomorphism

Definition

Let $h: G \to G'$ be a homomorphism of groups. The subgroup $h^{-1}[\{e'\}]=\{x \in G \mid h(x)=e'\}$ is the **kernel** of h, denoted by Ker(h).

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MTH633 Group Theory

Question No : 33 of 33

Let H and K be normal subgroups of a group G with $K \leq H$

Then show that $G/H \cong (G/K)/(H/K)$

Answer (Please click here to Add Answer)

Third Isomorphism Theorem

Theorem

Let H and K be normal subgroups of a group G with $K \leq H$.

Then $G/H \cong (G/K)/(H/K)$.

Third Isomorphism Theorem

Proof

Let $\phi: G \to (G/K)/(H/K)$ be given by $\phi(a) = (aK)(H/K)$ for $a \in G$.

Clearly ϕ is onto $(G/K)/(H/K)$, and for $a, b \in G$,

$\phi(ab) = [(ab)K](H/K)$

$= [(aK)(bK)](H/K)$

$= [(aK)(H/K)][(bK)(H/K)] = \phi(a)\,\phi(b)$,

so ϕ is a homomorphism.

Third Isomorphism Theorem

The kernel consists of those $x \in G$ such that $\phi(x) = H/K$.

These x are just the elements of H.

Then first isomorphism theorem shows that $G/H \cong (G/K)/(H/K)$.

Group Theory

Third Isomorphism Theorem

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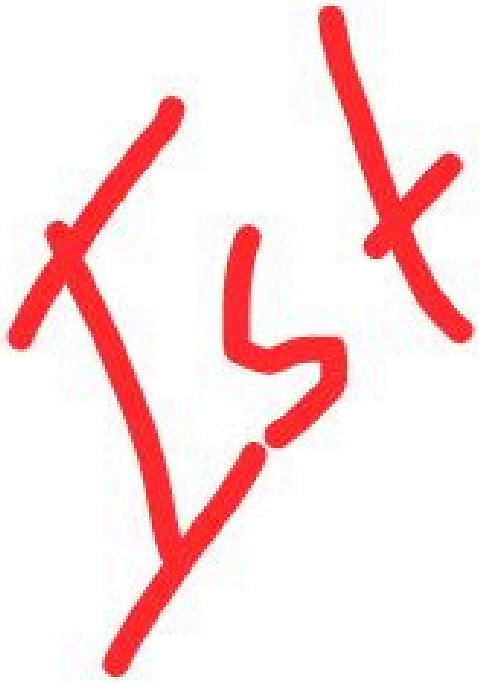
MTH633 Group Theory

Question No : 28 of 33

State the First theorem of Isomorphism.

Answer (Please click here to Add Answer)

Isomorphism Theorems



Theorem

Let $\phi: G \to G'$ be a homomorphism with kernel K, and let

$\gamma_K: G \to G/K$ be the canonical homomorphism. There is a unique isomorphism

$\mu: G/K \to \phi[G]$ such that $\phi(x) = \mu(\gamma_K(x))$ for each $x \in G$.

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MTH633 Group Theory

Question No : 33 of 33

Show that Z(G) is a normal and an abelian subgroup of G.

Answer (Please click here to Add Answer)

100%

ter Subgroup

Solution

For each $g \in G$ and

$z \in Z(G)$ we have

$gzg^{-1}=zgg^{-1}=ze=z$, we see at once that $Z(G)$ is a normal subgroup of G. It implies that $gz=zg$ for $g \in G$ and $z \in Z(G)$.

The Center Subgroup

Next

If G is abelian, then

Z(G) = G;

in this case, the center is not useful.

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MTH633 Group Theory

Question No : 30 of 33

If N is a normal subgroup of G, then show that G / N is abelian if and only if $C \leq N$, where C is commutator subgroup

Answer (Please click here to Add Answer)

Group Theory

The Commutator Subgroup

The Commutator Subgroup

Theorem

If N is a normal subgroup of G, then G/N is abelian if and only if $C \leq N$.

e Commutator Subgroup

Proof

If N is a normal subgroup of G and G/N is abelian, then

$(a^{-1}N)(b^{-1}N)=(b^{-1}N)(a^{-1}N)$; that is, $aba^{-1}b^{-1}N=N$,

so $aba^{-1}b^{-1} \in N$, and

$C \leq N$.

The Commutator Subgroup

Finally, if $C \leq N$, then

$(aN)(bN)=abN$

$=ab(b^{-1}a^{-1}ba)N$

$= (abb^{-1}a^{-1})baN$

$= baN$

$= (bN)(aN)$.